

Zarif Masud

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Research Interests

Responsible AI, algorithmic bias, information retrieval, migration studies, computational social science

Education

University of Toronto, Toronto, Canada PhD in Information, Faculty of Information Supervisors: Dr. Ebrahim Bagheri and Dr. Syed Ishtiaque Ahmed	Sep 2026 - Present
Toronto Metropolitan University, Toronto, Canada Doctoral studies in Computer Engineering	Sep 2023 - 2026
University of Waterloo, Waterloo, Canada MMath in Computer Science	Sep 2017 - Oct 2019
University of Dhaka, Dhaka, Bangladesh BSc in Software Engineering	Jan 2012 - Feb 2016

Experience

Researcher, BARTA - Remote, Dhaka, Bangladesh Mentor interdisciplinary teams on low-resource language processing and responsible AI projects.	Jul 2022 - Present
Software Engineer, Modern Data - Remote, Ontario, Canada Developed data engineering and analytics pipelines using Python, SQL, and AWS for enterprise clients.	Apr 2023 - Oct 2024
Software Development and Design Engineer, YuJa - Toronto, Canada Led the Himalayas project to reduce cloud storage costs for YuJa's Enterprise Video Platform by 60-70%.	Sep 2020 - Aug 2022
Associate Software Engineer, Kaz Software Ltd. - Dhaka, Bangladesh Built a full-stack information system for NGOs.	Jul 2016 - Jul 2017

Research Projects

Observatory on Immigration Discourses (IDIO), Bridging Divides Team member on an AI-assisted research observatory studying immigration discourse. Project page	Active
MS MARCO Immigration Narrative Bias Released data and reproducible analysis accompanying the ECIR 2026 paper. Research archive	2026

Publications

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- Emotional and Informational Trajectories of Immigrants: A Longitudinal Study of Reddit Communities** 2027
Zarif Masud, Abhijit Paul, Naimul Khan, Syed Ishtiaque Ahmed, and Ebrahim Bagheri.
Accepted for the International AAAI Conference on Web and Social Media (ICWSM 2027).
- The Role of Artificial Intelligence in Immigration: From Automation to Accountability** 2026
Zarif Masud, Syed Ishtiaque Ahmed, and Ebrahim Bagheri.
In Ana Beduschi (Ed.), *Handbook on Migration and Artificial Intelligence*. Edward Elgar Publishing.
- How Information Retrieval Systems Construct and Amplify Immigration Narratives** 2026
Zarif Masud, Abhijit Paul, Syed Ishtiaque Ahmed, and Ebrahim Bagheri.
Advances in Information Retrieval (ECIR 2026), pp. 474-488. DOI: 10.1007/978-3-032-21324-2_36.
- Partisan Perspectives on AI and Immigration: An Analysis of Canadian Parliamentary Discourse (2014-2024)** 2025
Eleyan Sawafta, Zarif Masud, Yasmeen Abu-Laban, Syed Ishtiaque Ahmed, Ebrahim Bagheri, and Geoffrey Rockwell.
Canadian Ethnic Studies, 57(3), 49-75. DOI: 10.1353/ces.2025.a989105.
- Development of a COVID-19-Related Anti-Asian Tweet Dataset: A Quantitative Study** 2023
Mahdi Mokhberi, Zarif Masud, Shebuti Rayana, and Syed Ishtiaque Ahmed.
JMIR Formative Research, 7(1), e40403. DOI: 10.2196/40403.
- SAFFRON: A Semi-Automated Framework for Software Requirements Prioritization** 2017
S. A. Asif, Zarif Masud, R. Easmin, and A. Gias.
International Journal of Advanced Computer Science and Applications, 8(12).

Workshop Papers

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- Exploring Cross-Lingual Knowledge Transfer via Transliteration-Based MLM Fine-Tuning for Critically Low-resource Chakma Language** 2025
Adity Khisa, Nusrat Jahan Lia, Tasnim Mahfuz Nafis, Zarif Masud, Tanzir Pial, Shebuti Rayana, and Ahmedul Kabir.
Proceedings of the Second Workshop on Bangla Language Processing (BLP-2025), pp. 280-291. DOI: 10.18653/v1/2025.banglalp-1.23.
- Sinophobia on Twitter: Integrating Theory, Humans, and Machines** 2021
Ahana Biswas, Zarif Masud, Shebuti Rayana, and Syed Ishtiaque Ahmed.
ACM CSCW Workshop on Addressing Challenges and Opportunities in Online Extremism Research.

Preprints

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- State-of-the-Art Translation of Text-to-Gloss Using mBART: A Case Study of Bangla** 2025
Sharif Mohammad Abdullah, Abhijit Paul, Zarif Masud, Shebuti Rayana, Abu Nayeem Md Touhidul Alam, Faisal Muhammad Shah, Md Shadab Iftikhar, Md Arid Hasan, and Ahmed Imtiaz Humayun.
arXiv:2504.02293.
- Where Journalism Silenced Voices: Indigenous Communities in Bangladeshi Media** 2025
Abhijit Paul, Arifur Rahman Khisa, Zarif Masud, Sharif Mohammad Abdullah, Arif Kabir, and Shebuti Rayana.
arXiv:2506.09771.
- Stemming: The Evolution and Current State with a Focus on Bangla** 2025
Abhijit Paul, Md Abu Farin, Zarif Masud, Sharif Mohammad Abdullah, Arif Kabir, and Saiful Islam.
arXiv:2508.15711.

Thesis

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- Switching GAN-Based Image Filters to Improve Perception for Autonomous Driving** 2019
Zarif Masud.
Master's thesis, University of Waterloo.